or of elastic waves in a dispersive medium. The signal in B can then be recorded with a pen connected to a buoy or with a microphone, respectively. As such, the signal will be detected as soon as the amplitude of the incoming wave has risen above the sensitivity threshold of our instrument. The information propagation speed is not necessarily the group velocity under these circumstances.

The information propagation speed may also be different from the group velocity in Eq. (4.6) for a square-law detector if the waveform is different from a wave packet. Take, for example, a "beginning sine", the ideal case of a wave that starts its existence at $t = 0$ and then continues as a sine forever. In this case, a square-law detector in B will not necessarily detect the front of the incoming wave, which is altered by dispersion, as having moved at the group velocity. As a matter of fact, the group velocity of light in a dispersive medium may be larger than the light velocity in a vacuum, as we shall see in the subsequent sections. This is quite an exceptional case, but historically, it raised doubts about a possible violation of the relativity principle. However, it was soon shown that, under those circumstances, the information propagation speed is different from group velocity and is, in any case, smaller than c.

Another similar example is the ideal case of a "finishing sinusoid", in which a sinusoidal wave that always existed suddenly ends at $t = 0$. We shall see later in this section how the speed of the end of the waves, as opposed to the above quoted exception, is equal to the group velocity for the surface waves on water.

The most familiar example of wavy phenomena is undoubtedly the waves on water surfaces. They are strongly dispersive and give us the opportunity to discuss the concepts we have introduced in concrete examples. Note, however, that a mathematical description of the surface waves is not at all simple, and we shall not enter into the demonstrations. Let us look at the principal characteristics.

First of all, the surface waves are neither transversal nor longitudinal. Let us fix our attention on a particle of water on the surface (or immediately below the surface) and on its trajectory when the wave passes through. The motion is neither an oscillation on a line perpendicular to the wave velocity, nor one on a line in the velocity direction. Looking carefully, we see that the water particle trajectory is an ellipses. This is simply a consequence of water being incompressible. If, at a certain point, the water surface should rise because a wave ridge is forming, more water has to move there, in practice, from a point in front, where a gorge is forming.

Secondly, as we anticipated, the surface waves are dispersive. The dispersion relation depends on the restoring forces acting on the water particle when it is out of equilibrium. These are the forces that make water surfaces flat at equilibrium (considering a dimension much smaller than the earth's radius). As a matter of fact, there are two restoring forces: the weight and the surface tension. Surface tension is important for the small ripples having wavelengths on the order of a millimeter (waves in a coffee pot). For larger wavelengths, the weight completely dominates. We shall only discuss the latter conditions here.

Thirdly, effects that are quite often dissipative are present. For example, in shallow water, the wave motion can reach the bottom of the basin, and here the friction with the ground can subtract energy from the wave. This is what happens

when a wave approaches a beach. We see that when the water depth decreases enough, the crest of the wave overturns and the wave breaks. Under these conditions, the system is not linear.

We shall limit our discussion to the gravity waves on the surface of an ideal liquid, which is incompressible (constant density ρ) and inviscous (viscosity $\eta = 0$).

Let h be the depth of the basin, namely the distance between the surface and the bottom. Observing the motion, we can state that the elliptic trajectories of the water particles have decreasing diameters with increasing depth. At the depth of one wavelength, the amplitude of the motion is reduced to 0.2 % of the amplitude on the surface. In addition, it can be shown that the dispersion relation is

$$\omega^2 = kg \tanh(kh), \tag{4.7}$$

where g is the gravity acceleration. Clearly, harmonic waves of different wavelengths have different wave velocities. We draw the attention of the reader to the argument of the hyperbolic tangent function, which is $kh = 2\pi h/\lambda$. It is proportional to the ratio of two lengths, the depth of the basin and the wavelength of the wave. There are two interesting limit cases, in which one of the two lengths is much greater than the other.

When $h \gg \lambda$, we speak of a deep water wave. In this case, considering that $\lim_{h\to\infty} \tanh kh = 1$, we have

$$\omega^2 = kg. \tag{4.8}$$

The phase velocity is then

$$v_p = \frac{\omega}{k} = \sqrt{\frac{g\lambda}{2\pi}}. \tag{4.9}$$

The group velocity is one half of the phase velocity, namely

$$v_g = \frac{d\omega}{dk} = \frac{1}{2} v_p. \tag{4.10}$$

We note that both velocities are independent of the depth of the water. This is because the motion of the water particles has already ceased at a depth on the order of a wavelength, as we already mentioned, and consequently, everything proceeds as if the bottom did not exist.

The phase velocity grows proportionally to the square root of the wavelength, namely longer waves travel faster than shorter ones. Figure 4.4 shows phase and group velocities as being functions of the wavelength. If, for example, a speedboat produces waves in the sea somewhat offshore, an observer on the beach will see the longer waves arriving first. Subsequently, waves of shorter wavelengths, and higher frequencies, will gradually appear.

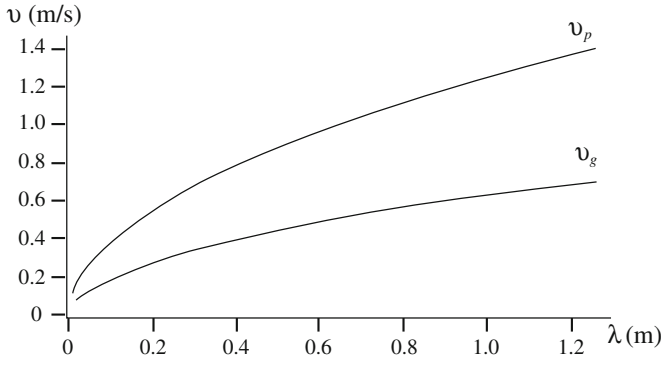


Fig. 4.4 Phase and group velocities of surface waves in deep water versus wavelength

QUESTION Q 4.1. What is the period of a sea wave of 10 m wavelength in deep water? And that of a wave of 2 m wavelength? What are their group and phase velocities? □

Let us now consider the opposite case of shallow water waves. We can now approximate $\tanh(kh) \approx kh$, and Eq. (4.7) becomes

$$\omega^2 = ghk^2. \tag{4.11}$$

Under these conditions, the system is non-dispersive, namely ω is proportional to k. Consequently, the phase and group velocities are equal to one another. They are given by

$$\upsilon_p = \upsilon_p = \sqrt{gh}. \tag{4.12}$$

The velocity increases with the square root of the depth.

The non-dispersive property of shallow water waves explains the disastrous phenomenon of the tsunamis originated by high magnitude quakes having their epicenter in the open sea. The seismic movement of the sea floor propagates to the entire water column up to the surface. The height of the column may be of a few kilometers. The resulting waves generally have small amplitude, only several centimeters in the open sea, but the mass of moving water is huge. The wavelengths are enormous, up to one hundred kilometers, namely so large that even the ocean is shallow water for them. As a consequence, these waves propagate without deformation, transporting all the original energy in a few crests. Their speed in the deep sea is very high, comparable to a sound wave in the atmosphere. Indeed, Eq. (4.12) gives us, for example, for a depth $h = 3$ km, $\upsilon_g = 170$ m/s. This corresponds, for a typical wavelength of 170 km, to a period of 1000 s. At this speed, if the epicenter is, say, at 10 km offshore, the waves will reach the seacoast in about 10'. When the wave is close to shore and the sea depth decreases, the height of the submerged liquid column and the speed of the wave decrease as well. In order for energy to be conserved, a large fraction of the kinetic energy transforms into potential energy and the height of the wave increases enormously. When the two or three crests hit the coastline, they sow destruction everywhere they hit.

Let us now schematically look at how we can measure the different velocities of the surface gravity waves in water. This will help us to understand how phase velocity, group velocity and energy (information) velocity are also different quantities when considered from the operational point of view.

Let us start by recalling the definitions. The phase velocity of a harmonic wave is

$$v_p = \frac{\omega}{k}, \tag{4.13}$$

which can be written in terms of the wavelength and the frequency as

$$v_p = \lambda \nu. \tag{4.14}$$

The group velocity is

$$v_g = \frac{d\omega}{dk}. \tag{4.15}$$

Suppose we perform our measurements in a canal of square cross-section, 1 m wide and 1 m deep, and some 20 m long. This is a wave-guide in which we can inject waves moving up and down a wooden septum on the surface at one extreme of the canal. In this way, we produce a wave group of a few crests separated by, say, a dozen centimeters and then study its propagation. We have prepared two reference lines forming a baseline that we have measured. We shall determine the velocities by measuring the time taken by the wave to travel through this base.

Let us start with the phase velocity. We can measure it by two methods.

With the first method, we fix our attention on a particular crest of the packet and measure the time it takes for it to go through the base. The phase velocity is obviously the ratio between the base length and this time.

With the second method, we take a photo of the packet at a certain instant and measure, on the picture, the average distance between crests. This is the average wavelength. We had also arranged a small buoy connected to a recorder that provides us with a record of the height of the surface as a function of time. From this record, we can easily extract the average oscillation frequency. Having measured wavelength and frequency, we multiply them to obtain the phase velocity.

The two methods are operationally different, but, as we can verify, give the same result within the experimental uncertainties.

We now measure the group velocity with a method similar to the first one above. However, we now look at the times of passage at the two lines of the maximum of the packet rather than of a single crest. In this way, we find that the group velocity is one half of the phase velocity.

We can finally measure the energy velocity by measuring the speed of the end of the waves. We inject into the guide a certain number of crests, fix our attention on the moving point at which the wave motion is just finished, and measure the time it takes to go through the base. We thus find the velocity with which energy propagates through the system. Note that, operationally, energy and group velocities are

different concepts. In this case, we experimentally find that energy and group velocities are equal to one another. However, this might not be the case for other systems.

4.2 Measurement of the Speed of Light

The word "light", strictly speaking, means the electromagnetic radiation in the frequency interval in which it can be "seen" by human eyes. In this section, we describe the first historical measurements of the speed of light. They were accomplished with astronomical observations, hence in vacuum, and subsequently in the atmosphere, in which the difference from a vacuum is, however, very small.

Let us start by recalling a few fundamental concepts that should be well known to the reader. The speed of light is extremely fast, or, more properly stated, it is the largest possible velocity. Indeed, as we discussed in the 1st volume of this course, the relativity principle, originally stated by Galileo Galilei (Italy, 1564–1642), holds for all laws of physics. We also recall that it can be shown that, under very general assumptions on the properties of space and time and the validity of the cause and effect principle, only two sets of transformations between the spatial and time coordinates of two inertial reference frames exist, namely the Galilei and the Lorentz transformations. The latter contain a constant, c, which has the physical dimensions of a velocity. The Galilei transformations are the limit of the Lorentz transformations for $c \to \infty$. The quantity c is one of the fundamental constants of physics, is invariant under Lorentz transformations, is the largest possible velocity, and is the velocity of the electromagnetic and gravitational waves in a vacuum. Given its importance, c was measured with precision increasing with time. The relative uncertainty was as small as four parts per billion ($\Delta c/c < 4 \times 10^{-9}$) in 1975, when the Conférence Générale de Poids et Mesures recommended redefining the unit of lengths, the meter, as the distance traveled by light in a vacuum in 1/299 792 458 of a second. Consequently, in the SI system of units, the value of c is fixed by definition to be equal to

$$c = 299\,792\,458 \text{ m s}^{-1}. \tag{4.16}$$

We finally recall that we learned in Sect. 3.6 that the wave equation in Eq. (3.31) is a solution to the Maxwell equations in a vacuum. Consequently, light waves in a vacuum are not dispersive and the phase and group velocities are equal.

Considering the extremely large value of the speed of light, it is not surprising that its propagation has been considered as instantaneous through the millennia of human culture. It was G. Galilei who first thought that light might propagate with a finite velocity and tried an experiment to measure it, as he later wrote in his "Two new sciences" published in 1638. The idea was as follows. Two people hold two lanterns, such that, by interposition of the hand, the one can shut off or admit the light to the vision of the other. They first practice, at a short distance from one